

DUSARI CHANDRA SHEKHAR

Backend Python Developer · Microservices & Distributed Systems · High-Performance APIs

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PROFESSIONAL SUMMARY

Backend Python Developer with **3+ years** of experience building scalable microservices, distributed systems, and high-performance APIs for high-traffic production environments. Skilled in FastAPI, Apache Kafka, Redis, Docker, and Kubernetes. Engineered systems processing **50,000+ daily requests** with 99.9% uptime, reduced API latency by **81%**, and shipped real-time WebSocket infrastructure for **5,000+ concurrent users**. Strong focus on scalability, performance optimization, and designing fault-tolerant distributed systems.

TECHNICAL SKILLS

Languages	Python, FastAPI, Flask
Architecture	Microservices, Distributed Systems, Event-Driven Architecture, System Design, Asynchronous Systems
Databases	MongoDB (Geospatial), PostgreSQL, MySQL, Redis
Messaging	Apache Kafka
DevOps & Cloud	Familiar With Docker, Kubernetes, AWS (EC2, S3), CI/CD (Jenkins, GitHub Actions)
API & Security	REST APIs, WebSockets, JWT, OAuth2, RBAC, Rate Limiting, Swagger/OpenAPI
Testing & Tools	Pytest, Postman, Git
Additional	Node.js, JavaScript, ReactJS Familiar With: Java, C++, PHP

WORK EXPERIENCE

Enmovil Solutions

Software Development Engineer · Hyderabad, India

- **Scaled logistics backend** to 50,000+ daily requests with 99.9% uptime across 12+ production services.
- Designed **12+ domain-driven microservices** using FastAPI & Flask — each independently deployable — for shipment tracking, warehouse coordination, and inventory management.
- Designed event-driven architecture with **Apache Kafka** to decouple services and enable scalable real-time GPS event streaming and shipment status propagation.
- Optimised **20+ high-throughput REST APIs** with Redis caching — reduced response time 800ms → 150ms (**81%**) and cut DB load significantly.
- Optimised MongoDB schemas with geospatial indexing and compound query tuning — reasoned through read/write trade-offs to achieve **65% faster** query response times.
- Implemented **Uber H3 hexagonal geospatial indexing** to replace costly polygon intersection queries — **40%+ latency reduction** on geofence lookups.
- Developed fault-tolerant async task pipelines for invoicing, alerts & reporting — handling **100,000+ daily jobs** with retry logic and dead-letter queues.
- Enforced layered API security: JWT, OAuth2, RBAC, rate limiting across all services; maintained **92% test coverage** via Pytest.
- Collaborated on Docker/Kubernetes deployments and CI/CD (Jenkins, GitHub Actions); mentored 3 junior developers on backend architecture patterns.

Wipro Limited

Associate Software Engineer · Hyderabad, India

Project: NetSuite — E-Commerce Payment Platform (PayPal Merchant Services)

- Built Python/Flask backend services for high-volume payment processing workflows.
- Developed REST APIs for payment gateway integrations, transaction history, and user management.
- Improved API response times by **30%** through MongoDB query optimisation and indexing.
- Created API test suites in Postman with **90%+ coverage**; produced full endpoint documentation.

PROJECTS

Real-Time Logistics Tracking System

FastAPI · Kafka · MongoDB · Redis · Docker

- Chose Kafka pub/sub over direct API calls to decouple GPS ingestion from downstream processors — eliminates cascade failures and enables independent scaling of each consumer service.
- Replaced polygon intersection queries with **Uber H3 hexagonal indexing** — reduced geofence lookup from $O(n)$ to $O(1)$, handling millions of location events/day at consistent latency.

Async Task Processing Platform

Python · Redis · Docker

- Designed with exponential back-off retries, dead-letter queues, and priority queues — time-sensitive alerts preempt bulk jobs; prevents invoice failures from blocking the request lifecycle. Handles **100,000+ daily tasks** with zero data loss.

EDUCATION

Bachelor of Technology — Computer Science & Engineering

Aditya Engineering College, Andhra Pradesh

2017 – 2021